

JARINGAN KOMPUTER / COMPUTERS NETWORK

SUBNETTING

STUDI KASUS

CASE STUDY

Sebuah perusahaan menerima alokasi jaringan 192.168.0.0/22 dari ISP. Perusahaan memiliki 4 departemen yang membutuhkan alokasi IP sebagai berikut:

A company has been assigned a network allocation of 192.168.0.0/22 from the Internet Service Provider (ISP). The company consists of four departments that require the following IP allocations:

Department	Number of Host
A	100
B	50
C	25
D	10

Tugas:

1. Buat perencanaan subnetting menggunakan metode VLSM
2. Buat perencanaan subnetting menggunakan metode FLSM
3. Tunjukkan bagaimana ISP menringkas alamat ke dalam satu entri routing menggunakan CIDR

Task:

1. Create a subnetting plan using the Variable Length Subnet Mask (VLSM) method
2. Create a subnetting plan using the Fixed Length Subnet Mask (FLSM) method
3. Demonstrate how the ISP can summarize the given network address into a single routing entry using Classless Inter-Domain Routing (CIDR)

A. VLSM

1. Langkah 1: Urutkan kebutuhan dari terbesar

Subnet	Host
A	100
B	50
C	25
D	10

Create by STO

2. Langkah 2: Tentukan ukuran blok masing-masing

Gunakan rumus: $2^n - 2 \geq$ jumlah host untuk ukuran blok
sedangkan subnet mask gunakan rumus: $32 - n$

- Subnet A

Host dibutuhkan: 100

Cari n: $2^6 = 64 - 2 < 100$ (belum memenuhi)

$2^7 = 128 - 2 \geq 100$ (memenuhi)

Maka: Allocated Size = 128 dan Assignable Host = 126

n = 7

Subnet Mask: $32 - 7 = 25$ sehingga Decimal mask:

1111 1111.1111 1111.1111 1111.1000 0000 (terdapat

25 bits 1) yang dalam decimal menjadi

255.255.255.128

- Subnet B

Host dibutuhkan: 50

Cari n: $2^5 = 32 - 2 < 50$ (belum memenuhi)

$2^6 = 64 - 2 \geq 50$ (memenuhi)

Maka: Allocated Size = 64 dan Assignable Host = 62

n = 6

Subnet Mask: $32 - 6 = 26$ sehingga Decimal mask:

1111 1111.1111 1111.1111 1111.1100 0000 (terdapat

26 bits 1) yang dalam decimal menjadi

255.255.255.192

- Subnet C

Host dibutuhkan: 25

Cari n: $2^4 = 16 - 2 < 25$ (belum memenuhi)

$2^5 = 32 - 2 \geq 25$ (memenuhi)

Maka: Allocated Size = 32 dan Assignable Host = 30

n = 5

Subnet Mask: $32 - 5 = 27$ sehingga Decimal mask:

1111 1111.1111 1111.1111 1111.1110 0000 (terdapat

27 bits 1) yang dalam decimal menjadi

255.255.255.224

- Subnet D

Host dibutuhkan: 10

Cari n: $2^3 = 8 - 2 < 10$ (belum memenuhi)

$2^4 = 16 - 2 \geq 10$ (memenuhi)

Maka: Allocated Size = 16 dan Assignable Host = 14

n = 4

Subnet Mask: $32 - 4 = 28$ sehingga Decimal mask:

1111 1111.1111 1111.1111 1111.1111 0000 (terdapat

28 bits 1) yang dalam decimal menjadi

255.255.255.240

3. Langkah 3: Alokasi IP Address

- Alokasi Subnet A:
 Alamat awal: 192.168.0.0
 Ukuran blok: 128 IP → 0-127
 Broadcast: 192.168.0.127
 Assignable range: 192.168.0.1 - 192.168.0.126 (126)
- Alokasi Subnet B:
 Alamat awal berikutnya: 192.168.0.128
 Ukuran blok: 64 → 128-191
 Broadcast: 192.168.0.191
 Assignable: 192.168.0.129 - 192.168.0.190 (62)
- Alokasi Subnet C:
 Alamat awal: 192.168.0.192
 Ukuran blok: 32 → 192-223
 Broadcast: 192.168.0.223
 Assignable: 192.168.0.193 - 192.168.0.222 (30)
- Alokasi Subnet D:
 Alamat awal: 192.168.0.224
 Ukuran blok: 16 → 224-239
 Broadcast: 192.168.0.239
 Assignable: 192.168.0.225 - 192.168.0.238 (14)

Tabel Akhir

Subnet	Needed Host	Allocated Size	Address	Mask	Decimal Mask	Assignable Range	Broadcast
A	100	128	192.168.0.0/25	/25	255.255.255.128	192.168.0.1 - 192.168.0.126	192.168.0.127
B	50	64	192.168.0.128/26	/26	255.255.255.192	192.168.0.129 - 192.168.0.190	192.168.0.191
C	25	32	192.168.0.192/27	/27	255.255.255.224	192.168.0.193 - 192.168.0.222	192.168.0.223
D	10	16	192.168.0.224/28	/28	255.255.255.240	192.168.0.225 - 192.168.0.238	192.168.0.239

1. Step 1: Sort the requirements from largest to smallest

Subnet	Host
A	100
B	50
C	25
D	10

2. Step 2: Determine the block size for each subnet

- Subnet A
 Required number of hosts: 100
 Determine n:
 $2^6 = 64 - 2 < 100 \rightarrow$ does not satisfy the requirement
 $2^7 = 128 - 2 \geq 100 \rightarrow$ satisfies the requirement

Therefore:

Allocated Block Size = 128
Usable Hosts per Subnet = 126
Host Bits Required (n) = 7
Subnet Mask = /25 (since $32 - 7 = 25$)
Binary Subnet Mask:
11111111.11111111.11111111.10000000 (25 bits set to 1)
Decimal Subnet Mask: 255.255.255.128

- Subnet B

Required number of hosts: 50

Determine n:

$2^5 = 32 - 2 < 50 \rightarrow$ does not satisfy the requirement

$2^6 = 64 - 2 \geq 50 \rightarrow$ satisfies the requirement

Therefore:

Allocated Block Size = 64
Usable Hosts per Subnet = 62
Host Bits Required (n) = 6
Subnet Mask = /26 (since $32 - 6 = 26$)
Binary Subnet Mask:
11111111.11111111.11111111.11000000 (26 bits set to 1)
Decimal Subnet Mask: 255.255.255.192

- Subnet C

Required number of hosts: 25

Determine n:

$2^4 = 16 - 2 < 25 \rightarrow$ does not satisfy the requirement

$2^5 = 32 - 2 \geq 25 \rightarrow$ satisfies the requirement

Therefore:

Allocated Block Size = 32
Usable Hosts per Subnet = 30
Host Bits Required (n) = 5
Subnet Mask = /27 (since $32 - 5 = 27$)
Binary Subnet Mask:
11111111.11111111.11111111.11100000 (27 bits set to 1)
Decimal Subnet Mask: 255.255.255.224

- Subnet D

Required number of hosts: 10

Determine n:

$2^3 = 8 - 2 < 10 \rightarrow$ does not satisfy the requirement

$2^4 = 16 - 2 \geq 10 \rightarrow$ satisfies the requirement

Therefore:

Allocated Block Size = 16
Usable Hosts per Subnet = 14
Host Bits Required (n) = 4
Subnet Mask = /28 (since $32 - 4 = 28$)

Binary Subnet Mask:

11111111.11111111.11111111.11110000 (28 bits set to 1)

Decimal Subnet Mask: 255.255.255.240

3. Step 3: Allocate the IP addresses

- Subnet A:
Starting Address: 192.168.0.0
Block Size: 128 IPs → Range: 0-127
Broadcast Address: 192.168.0.127
Assignable Host Range: 192.168.0.1 – 192.168.0.126 (126 usable hosts)
- Subnet B:
Starting Address: 192.168.0.128
Block Size: 64 IPs → Range: 128-191
Broadcast Address: 192.168.0.191
Assignable Host Range: 192.168.0.129 – 192.168.0.190 (62 usable hosts)
- Subnet C:
Starting Address: 192.168.0.192
Block Size: 32 IPs → Range: 192-223
Broadcast Address: 192.168.0.223
Assignable Host Range: 192.168.0.193 – 192.168.0.222 (30 usable hosts)
- Subnet D:
Starting Address: 192.168.0.224
Block Size: 16 IPs → Range: 224-239
Broadcast Address: 192.168.0.239
Assignable Host Range: 192.168.0.225 – 192.168.0.238 (14 usable hosts)

Final Table

Subnet	Needed Host	Allocated Size	Address	Mask	Decimal Mask	Assignable Range	Broadcast
A	100	128	192.168.0.0/25	/25	255.255.255.128	192.168.0.1 – 192.168.0.126	192.168.0.127
B	50	64	192.168.0.128/26	/26	255.255.255.192	192.168.0.129 – 192.168.0.190	192.168.0.191
C	25	32	192.168.0.192/27	/27	255.255.255.224	192.168.0.193 – 192.168.0.222	192.168.0.223
D	10	16	192.168.0.224/28	/28	255.255.255.240	192.168.0.225 – 192.168.0.238	192.168.0.239

B. FLISM

1. Langkah 1: Tentukan Kebutuhan Terbesar

Diketahui Subnet A membutuhkan 100 host, maka seluruh subnet harus dialokasikan ukuran yang cukup untuk menampung minimal 100 host sehingga

$$2^7 = 128 - 2 \geq 100 \text{ (memenuhi)}$$

Create by STO

Maka: Allocated Size = 128 dan Assignable Host = 126
 $n = 7$
 Subnet Mask: $32 - 7 = 25$ sehingga Decimal mask:
 1111 1111.1111 1111.1111 1111.1000 0000 (terdapat
 25 bits 1) yang dalam decimal menjadi
 255.255.255.128

2. Langkah 2: Alokasi IP Address

Mulai dari 192.168.0.0, kita bagi jaringan 192.168.0.0/22 menjadi blok-blok /25 (128 IP).

Total IP dalam /22 = 1024

Jumlah blok /25 = $1024/128 = 8$ blok tersedia → cukup

Alokasi Subnet A:

Alamat awal: 192.168.0.0

Ukuran blok: 128 IP → 0-127

Broadcast: 192.168.0.127

Assignable range: 192.168.0.1 - 192.168.0.126

Alokasi Subnet B:

Alamat awal berikutnya: 192.168.0.128

Blok: 128 → 128-255

Broadcast: 192.168.0.255

Assignable: 192.168.0.129 - 192.168.0.254

Alokasi Subnet C:

Alamat awal: 192.168.1.0 (lanjutan dari 192.168.0.255 + 1)

Blok: 128 → 192.168.1.0-192.168.1.127

Broadcast: 192.168.1.127

Assignable: 192.168.1.1 - 192.168.1.126

Alokasi Subnet D:

Alamat awal: 192.168.1.128

Blok: 128 → 192.168.1.128-255

Broadcast: 192.168.1.255

Assignable: 192.168.1.129 - 192.168.1.254

Tabel Akhir

Subnet	Needed Host	Allocated Size	Address	Mask	Decimal Mask	Assignable Range	Broadcast
A	100	128	192.168.0.0/25	/25	255.255.255.128	192.168.0.1 - 192.168.0.126	192.168.0.127
B	50	128	192.168.0.128/25	/25	255.255.255.128	192.168.0.129 - 192.168.0.254	192.168.0.255
C	25	128	192.168.1.0/25	/25	255.255.255.128	192.168.1.1 - 192.168.1.126	192.168.1.127
D	10	128	192.168.1.128/25	/25	255.255.255.128	192.168.1.129 - 192.168.1.254	192.168.1.255

1. Step 1: Identify the Largest Requirement

Given that Subnet A requires 100 hosts, all subnets must be allocated with a block size sufficient to accommodate at least 100 usable host addresses.

To determine the appropriate block size:

$2^7 = 128$ total addresses

Subtracting 2 for the network and broadcast addresses gives 126 usable host addresses

Since $126 \geq 100$, this block size satisfies the requirement

Therefore:

Allocated Block Size: 128

Usable Host Addresses: 126

Host Bits Required (n): 7

Subnet Mask: /25 (since $32 - 7 = 25$)

Binary Representation of Subnet Mask:

11111111.11111111.11111111.10000000 (25 bits set to 1)

Decimal Representation of Subnet Mask: 255.255.255.128

2. Step 2: Allocate IP Addresses

Starting from the network address 192.168.0.0/22, we divide the network into blocks of /25 (each block contains 128 IP addresses).

Total number of usable IP addresses in /22 network:

$2^{(32-22)} = 1024$ IP addresses

Number of /25 subnets available: $1024/128 = 8$ subnets → sufficient for all four departments using FLSM.

Subnet Allocation Using FLSM (/25 per Subnet)

Subnet A:

Starting Address: 192.168.0.0

Block Size: 128 IPs → Range: 192.168.0.0 - 192.168.0.127

Broadcast Address: 192.168.0.127

Usable Host Range: 192.168.0.1 - 192.168.0.126

Subnet B:

Starting Address: 192.168.0.128

Block Size: 128 IPs → Range: 192.168.0.128 - 192.168.0.255

Broadcast Address: 192.168.0.255

Usable Host Range: 192.168.0.129 - 192.168.0.254

Subnet C:

Starting Address: 192.168.1.0 (continuation from previous block)

Block Size: 128 IPs → Range: 192.168.1.0 - 192.168.1.127
 Broadcast Address: 192.168.1.127
 Usable Host Range: 192.168.1.1 - 192.168.1.126

Subnet D:

Starting Address: 192.168.1.128
 Block Size: 128 IPs → Range: 192.168.1.128 - 192.168.1.255
 Broadcast Address: 192.168.1.255
 Usable Host Range: 192.168.1.129 - 192.168.1.254

Final Table

Subnet	Needed Host	Allocated Size	Address	Mask	Decimal Mask	Assignable Range	Broadcast
A	100	128	192.168.0.0/25	/25	255.255.255.128	192.168.0.1 - 192.168.0.126	192.168.0.127
B	50	128	192.168.0.128/25	/25	255.255.255.128	192.168.0.129 - 192.168.0.254	192.168.0.255
C	25	128	192.168.1.0/25	/25	255.255.255.128	192.168.1.1 - 192.168.1.126	192.168.1.127
D	10	128	192.168.1.128/25	/25	255.255.255.128	192.168.1.129 - 192.168.1.254	192.168.1.255

C. CIDR

Blok gabungan: 192.168.0.0/22
 Rentang IP: 192.168.0.0 - 192.168.3.255
 Jumlah IP: $2^{32-22} = 1024$ IP

Tabel Akhir

CIDR Block	Network Range	Total Network	Total IP	Assignable Host IP
192.168.0.0/22	192.168.0.0 - 192.168.3.255	4 × /24	1024	192.168.0.1 - 192.168.3.254
Subnet Mask	255.255.252.0			

Combined Network Block: 192.168.0.0/22
 IP Address Range: 192.168.0.0 - 192.168.3.255
 Total Number of IP Addresses: $2^{(32-22)} = 1024$ IP addresses

Final Table

CIDR Block	Network Range	Total Network	Total IP	Assignable Host IP
192.168.0.0/22	192.168.0.0 - 192.168.3.255	4 × /24	1024	192.168.0.1 - 192.168.3.254
Subnet Mask	255.255.252.0			